

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 1st Year 1st Term Examination, 2023

Math 1113

(Differential and Integral Calculus)

Time: 3 Hours.

Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION – A

- 1(a). Define continuity and differentiability of a function. A function $f(x)$ is defined as follows: 14

$$f(x) = 2 - x \text{ for } 1 \leq x \leq 2$$

$$= x - \frac{1}{2}x^2 \text{ for } x > 2$$

Discuss the continuity and differentiability of $f(x)$ at $x=2$

- 1(b). State Taylor's theorem. Expand $f(x) = x^4 - 10x^3 + 3x^2 - 9x + 2$ in a Taylor's series in power $x - 2$. 10

- 1(c). Evaluate $\lim_{x \rightarrow 0} \left(\frac{\tan x}{x} \right)^{1/x}$ 11

- 2(a). Define tangent and normal. Find the equation of the tangent and normal to the curve $y = x^3 - 2x^2 - x + 5$ at the point (0,5) 12

- 2(b). If $\sin y = x \sin(a + y)$, then prove that $\frac{dy}{dx} = \frac{\sin^2(a+y)}{\sin a}$ 10

- 2(c). Define asymptotes. Find the asymptotes of the cubic $x^3 + x^2y - xy^2 - y^3 + 2xy + 2y^2 - 3x + y = 0$. 13

- 3(a). Find the maxima and minima of $1 + 2 \sin x + 3 \cos^2 x$, $0 \leq x \leq \frac{1}{2}\pi$ 12

- 3(b). State Leibnitz's theorem. If $y = (\sin^{-1} x)^2$, then prove that 13

(i) $(1 - x^2)y^2 - xy_1 - 2 = 0$

(ii) $(1 - x^2)y_{n+2} - (2n+1)xy_{n+1} - n^2y_n = 0$

- 3(c). Find y_n where $y = \frac{1}{x^2 + x + 1}$ 10

- 4(a). State Euler's theorem. If $u = \cos^{-1} \left\{ \frac{(x+y)}{(\sqrt{x} + \sqrt{y})} \right\}$, then show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + \frac{1}{2} \cot u = 0$ 13

- 4(b). Define radius of curvature. Show that for the curve $r^m = a^m \cos m\theta$ the radius of curvature is $\zeta = \frac{a^m}{(m+1)r^{m-1}}$ 13

- 4(c). Define subtangent and subnormal. In the curve $x^m y^n = a^{m+n}$, then show that the subtangent at any point varies as the abscissa of the point. 11

SECTION – B

- 5(a). Integrate $\int e^{2x} \frac{1 + \sin 2x}{1 + \cos 2x} dx$ 11
- 5(b). Integrate $\int \frac{1}{3 \sin hx + 2 \cos hx + 5} dx$ 13
- 5(c). Integrate $\int (3x - 2)\sqrt{(x^2 - x + 1)} dx$ 11
- 6(a). Integrate $\int_8^{15} \frac{dx}{(x-3)\sqrt{(x+1)}}$ 11
- 6(b). Evaluate $\int_0^{\pi/4} (1 + \tan \theta) d\theta$ 11
- 6(c). State Wallis's theorem. If $I_n = \int_0^{\pi/2} x^n \cos x dx$, then show that $I_n = \left(\frac{\pi}{2}\right)^n - n(n-1)I_{n-2}$. 13
Hence evaluate $\int_0^{\pi/2} x^5 \cos x dx$.
- 7(a). Define Gamma and Beta functions using Gamma and/ or Beta function, evaluate 13
(i) $\int_0^{\pi/2} \sin^7 \theta \cos^3 \theta d\theta$
(ii) $\int_0^1 x^3 (1-x)^4 dx$
- 7(b). Prove that $\sqrt{\frac{1}{2}} = \sqrt{\pi}$ and $\int_0^{\infty} e^{-x^2} dx = \frac{\sqrt{\pi}}{2}$ 12
- 7(c). Find the area of a loop of the curve $r = a \cos 3\theta$. Hence find the whole area of the curve. 10
- 8(a). Show that the length of the arc of the curve $r = a(1 + \cos \theta)$ between the $\theta = 0$ to the point $\theta = \frac{\pi}{2}$ is $2a\sqrt{2}$ 11
- 8(b). Find the volume of the solid generated by the revolving about the x-axis of a loop of the curve $2ay^2 = x(x-a)^2$ 12
- 8(c). Find the volume and the surface-area of the solid generated by revolving the cycloid $x = a(\theta + \sin \theta)$, $y = a(1 + \cos \theta)$ about its base. 12

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 1st Year 1st Term Examination, 2023

Hum 1113

(Sociology and Behavioral Science)

Time: 3 Hours.

Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION – A

- | | | |
|-------|--|----|
| 1(a). | What is meant by Sociology? | 05 |
| 1(b). | Describe the origin and development of sociology. | 20 |
| 1(c). | Why studying sociology is important for the student of engineering? | 10 |
| 2(a). | What is community? What are the differences between society and community? | 15 |
| 2(b). | Define culture. Describe functions and characteristics of culture with examples. | 15 |
| 2(c). | What is meant by cultural lag? | 05 |
| 3(a). | What is meant by socialization? Narrate the agencies of socialization with examples. | 15 |
| 3(b). | Define family. Discuss the functions of family. | 15 |
| 3(c). | What is meant by social stratification? | 05 |
| 4(a). | What is urbanization? Explain the consequences of urbanization in the context of Bangladesh. | 15 |
| 4(b). | Define deviance. Distinguish between deviance and crime. | 15 |
| 4(c). | What is meant by social control? Explain with examples. | 05 |

SECTION – B

- | | | |
|-------|---|----|
| 5(a). | How does knowledge of behavioral science reflect in the personal, professional, and social life of engineering students? | 15 |
| 5(b). | Describe the basic elements considered in a training evaluation. | 10 |
| 5(c). | Demonstrate the ways of improving employee relation. | 10 |
| 6(a). | What is organization design? Explain the principles of organization design. | 20 |
| 6(b). | Discuss the common reasons of psychological basis that create resistance to change. | 15 |
| 7(a). | Do you think the merits of 'Quality of Life' outweigh the demerits of it? Write the arguments in favor of your position. | 15 |
| 7(b). | Show the ways of maintaining psychological contract at work. | 10 |
| 7(c). | Evaluate the significance of counseling at workplace. | 10 |
| 8(a). | What are the basic elements of leadership? Analyze the limitations of engineering students for achieving a leadership role in work and social life. | 20 |
| 8(b). | Describe the ways to improve the intergroup cooperation in an organization. | 15 |

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 1st Year 1st Term Examination, 2023

Ch 1113

(Chemistry I)

Time: 3 Hours.

Full Marks: 210

- N.B. i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

SECTION – A

- 1(a). Define reversible and irreversible reactions? What are the factors that influence on chemical equilibrium? 10
- 1(b). Establish the relation between K_p and K_c . Explain with the help of Le Chatelier's principle, the effect of pressure on the following reaction: 10
- $$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$
- 1(c). How does Le Chatelier's principle important for the synthesis of ammonia by Haber-Bosch process? 08
- 1(d). The equilibrium constant for the synthesis of NH_3 by Haber-Bosch process is 1.65×10^{-4} at 673 K. What will be the equilibrium constant at 773 K if the enthalpy of reaction is -105.20 kJ? 07
- 2(a). What is gas electrode? Construct a standard H_2 gas electrode. What are the drawbacks for using standard hydrogen electrode? 10
- 2(b). How does PEN fuel cell work? Explain. 09
- 2(c). What is electrical double layer? Draw three models of electrical double layer proposed by Helmholtz, Gouy-Chapmann and Stern. 10
- 2(d). Write the cell reactions and calculate the free energy of the cell. Also comment on it's spontaneity. 06
- $$Ag(s) | Ag^+(aq) || Cu^{2+}(aq) | Cu(s)$$
- Where, $E_{Cu^{2+}/Cu}^0 = 0.34 V$ and $E_{Ag^+/Ag}^0 = 0.80 V$
- 3(a). What is nuclear binding energy? " ^{56}Fe is the most stable of all nuclides" – Explain it with the nuclear binding energy curve. 10
- 3(b). What are endoergic and exoergic reactions? Schematically describe 'Liquid-drop' model of nuclear reaction. 09
- 3(c). Write the differences between the nuclear fission and the nuclear fusion process. 08
- 3(d). Write a note on Van de Graaff generator for particle accelerators. 08
- 4(a). What is Li-ion battery? Discuss charging and discharging mode of this battery. 12
- 4(b). Define electroplating. What is the basic principle of electroplating? 08
- 4(c). What is nuclear magic number? Describe 'Electromagnetic method' for the separation of radioisotopes. 09
- 4(d). How many modes of mass transfer are there? Define them. 06

SECTION – B

5(a).	Define the following terms with examples: (i) Base unit, (ii) Living polymer, (iii) Inorganic polymer.	09
5(b).	What is conducting polymer? How can conducting polymer be synthesized? Mention some applications of conducting polymer.	10
5(c).	What is chain terminating agent? Discuss the mechanism of anionic polymerization.	10
5(d).	The degree of polymerization of polyvinyl acetate is 10,000. Calculate the molecular weight of polyvinyl acetate.	06
6(a).	Justify the following facts: (i) Natural rubber is purely hydrocarbon, (ii) Natural rubber has one double bond in its monomeric unit.	08
6(b).	Make a classification of fibers. Sketch modulus vs temperature graph and show different regions.	12
6(c).	What is Doctor's treatment? Explain.	08
6(d).	"CNG is considered as safe fuel" – Why?	07
7(a).	What are the properties to assess the quality of a coal? Discuss with significance of them.	10
7(b).	What is petroleum? Illustrate the synthesize procedure of liquid fuel from solid fuel?	10
7(c).	Define octane number. How can octane number of a fuel be improved?	08
7(d).	Make a differentiate between coal, coke and charcoal.	07
8(a).	Discuss a method to determine dissolved oxygen with chemical reactions.	10
8(b).	What is acid rain? What are the roles of oxides of nitrogen and Sulphur on it?	10
8(c).	"Several heavy metals exert toxic effects on living systems" - Explain the mechanism of toxicity.	08
8(d).	Distinguish between BOD and COD.	07

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 1st Year 1st Term Examination, 2023

Ph 1113

(Physics)

Time: 3 Hours.

Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION – A

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|-------|---|----|
| 1(a). | What is quantum theory of light? Give an account of Einstein's explanation of photoelectric effect. | 10 |
| 1(b). | Discuss analytically the formation of wave group or wave packet. Show that the de Broglie wave group associated with a moving body travel as the velocity of the body. | 15 |
| 1(c). | An X-ray photon of wavelength of $0.35 \text{ \AA}$ undergoes a 62° Compton scattering. Find the wavelength of the scattered photon and the kinetic energy imparted to the recoil electron. | 10 |
| 2(a). | Explain Correspondence principle. Find out the origin of spectral lines for Hydrogen atom. | 10 |
| 2(b). | Explain the physical significance of the different quantum numbers used in the Vector atom model. Also give two important applications of this model. | 15 |
| 2(c). | (i) Find the frequencies of revolution of electrons in $n = 1$ and $n = 2$ Bohr orbits.
(ii) What is the frequency of the photon emitted when an electron in an $n = 2$ orbit drops to an $n = 1$ orbit? | 10 |
| 3(a). | What is Radioactivity? Discuss the law governing of successive disintegration and radioactive equilibrium with two possible cases. | 15 |
| 3(b). | Define binding energy and packing fraction. Draw the binding energy per nucleon curve and explain the nuclear fission and fusion process from this graph. | 12 |
| 3(c). | 1 gram of Radium is reduced to 2.2 milligram in 5 years by α -decay. Calculate the half life and mean life of Radium. | 08 |
| 4(a). | What is Mirror nuclei? Write down the general properties of the atomic nucleus. | 08 |
| 4(b). | Describe briefly the various components of a nuclear reactor. Explain the principle of its working. When is the reactor said to be critical? | 15 |
| 4(c). | What is nuclear fusion process? Describe the proton-proton cycle and Carbon-Nitrogen cycle of the production of stellar energy. | 12 |

SECTION – B

- 5(a). Define simple harmonic motion. Show that an LC circuit behaves as simple harmonic oscillator. 12
- 5(b). Suppose two simple harmonic motions with different phases, amplitudes and time periods ratio 1:2 superimpose perpendicularly over a particle. Find the general expression for the resultant vibration of the particle. Show that the resultant path is parabola when the phase difference is $\frac{\pi}{2}$. 13
- 5(c). The force and displacement of a simple dynamic system undergoing sinusoidal excitation are given by the following equations: 10
- $$F = 10 \sin\left(\frac{\pi t}{10}\right) \text{ newtons}$$
- $$y = 0.1 \sin\left(\frac{\pi t}{10} - \frac{\pi}{6}\right) \text{ metres}$$
- Calculate the work done by the excitation force in 2 minutes.
- 6(a). What is a plane progressive wave? Show that the energy density of a plane progressive wave is given by $E = 2\pi^2 n^2 a^2 \rho$, where the symbols have their usual meanings. 13
- 6(b). A particle executing damped harmonic oscillation is subjected to an external periodic force. Establish the differential equation for the motion of the particle. Solve the equation and find the amplitude of the forced oscillator. 12
- 6(c). The displacement (in metres) of a particle executing simple harmonic motion at any instant of time is given by: 10
- $$y = 0.1 \sin 2\pi(340t - 0.15)$$
- Calculate (i) the amplitude of the vibrating particle, (ii) wave velocity, (iii) wavelength, (iv) frequency, and (v) time period.
- 7(a). What is atomic packing factor? Show that the atomic packing factor for simple cubic, body centered cubic and face centered cubic lattices are $\frac{\pi}{6}$, $\frac{\pi\sqrt{3}}{8}$, and $\frac{\pi}{3\sqrt{2}}$, respectively. 13
- 7(b). What is Miller indices? Show that for a cubic lattice, the interplanar distance between a set of (hkl) planes is given by: 12
- $$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$
- Where a is the lattice parameter.
- 7(c). Sodium is a body centered cubic crystal. Its density is $9.6 \times 10^2 \text{ kg/m}^3$ and atomic weight is 23. Calculate the lattice constant for a sodium crystal. 10
- 8(a). Mention the outstanding properties of metals. Obtain an expression for thermal conductivity of metals according to classical free electron theory. 13
- 8(b). Discuss about the classical theory of lattice heat capacity. Find the success and limitation of this theory. 12
- 8(c). A strip of copper of $150 \mu\text{m}$ thick is placed in a magnetic field of magnitude of 0.65 T and a current 23 amp is sent through the strip. What potential difference will appear across the width of the strip? The number of charge carriers per unit volume of the copper is $8.49 \times 10^{28} \text{ m}^{-3}$. 10

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 1st Year 1st Term Examination, 2023

ESE 1101

(Fundamentals of Energy Resources)

Time: 3 Hours.

Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION – A

- | | | |
|-------|---|----|
| 1(a). | What is meant by energy resources? Briefly explain the advantages and disadvantages of non-conventional energy sources. | 14 |
| 1(b). | Briefly explain following terms:
(i) Primary Energy, (ii) Secondary Energy, (iii) Commercial Energy | 06 |
| 1(c). | What is meant by energy cycle? Briefly explain the flow of energy in the earth-energy cycle with neat sketch. | 15 |
| 2(a). | Define nuclear energy. Explain the principle of nuclear power generator on with a schematic diagram. | 10 |
| 2(b). | Differentiate between nuclear fission and fusion. How does energy gets liberated in nuclear reaction? What is meant by nuclear fuel enrichment? | 12 |
| 2(c). | Write a brief note on sources, reserve and and classification of natural gas. | 08 |
| 2(d). | What are the environmental impacts of nuclear energy production? | 05 |
| 3(a). | Describe Abiogenic and Carbide therory for the origin of petroleum. What are the flaws of Carbide theory? | 09 |
| 3(b). | Describe the composition of natural gas. Write short note on different form of natural gas. | 10 |
| 3(c). | Explain in brief the extraction of petroleum oil. Compare primary oil recovery and secondary oil recovery. | 09 |
| 3(d). | What are the physical properties of natural gas? | 07 |
| 4(a). | How coal is formed? Expalin the metamorphosis of peat to coal. | 10 |
| 4(b). | Write down the schematic diagram of coal genesis. | 10 |
| 4(c). | Describe the advantages and disadvantages of using coal to produce electricity. | 08 |
| 4(d). | Write is better coal or coke ? Write down the application of coke. | 07 |

SECTION – B

5(a).	Which reactions are responsible for liberating energy within the sun? How solar radiation is measured on earth surface?	10
5(b).	How solar collectors are classified? Describe with neat sketch the construction of FPC.	09
5(c).	How a photovoltaic cell works? Explain with neat sketch. What is meant by dopping?	08
5(d).	With a neat sketch describe how solar energy can be used for desalination of water?	08
6(a).	Define bio-energy. How is it formed? Why biomass is need to be converted?	09
6(b).	What is meant by anaerobic digestion process? Explain how biogas is obtained through this process.	10
6(c).	How liquid fuel could be obtained from biomass? Describe how fermentation is occurred for non-sugar materials.	08
6(d).	What are the main components of a biogas reactor? What factors affect biogas production.	08
7(a).	Show schematically how hydro power plant works? With neat sketch show the classification of hydropower plant based on constructional feature. Also explain in brief.	09
7(b).	Write a brief note on main types of hydro turbines.	08
7(c).	How wind power could be exploited? Show with schematic diagram the major categories of wind turbines.?	09
7(d).	What factors are important for the selection of wind energy conversion system. Explain in brief.	09
8(a).	What are the different types of energy sources form Ocean? Describe the system concept of OTEC. What are the main components of OTEC?	10
8(b).	Explain how continuous generation of electricity is obtainable from a tidal power scheme.	09
8(c).	Write in brief the environmental aspects of wave energy with schematic of coastline alternation.	08
8(d).	Briefly explain the environmental effect of geothermal energy extraction.	08

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